

## UNDERSPECIFIED PLAN

```

sequential([
  a(PickUpAction)(
    arm=LEFT,
    object_designator=

    a(DetectedMontessoriShape)(
      category=CUBE)
    .where(
      Colored(shape, CYAN),
      SupportedBy(shape, lid)),

  grasp_description=

  a(GraspDescription)(
    approach_direction=...,
    vertical_alignment=TOP,
    end_effector=LEFT_HAND)

),
an(InsertionAction)(
  arm=LEFT,
  object_designator=shape,
  grasp_description=grasp,
  target=

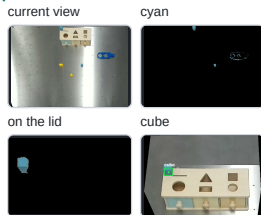
  a(ShapeSortingHole)(
    shape_category=...)
    .from_(board.apertures)

])

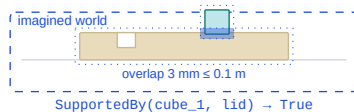
```

## RESOLVED BY

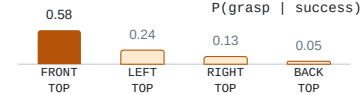
### PerceptionBackend



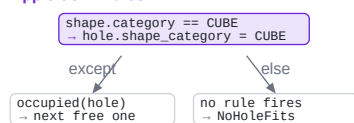
### Simulation backend



### ProbabilisticBackend



### Ripple-down rules



## RESOLVED PLAN

```

sequential([
  PickUpAction(
    arm=LEFT,
    object_designator=

    cube_1 # CYAN, CUBE
    at (0.72, 0.13, 0.10) m,
    SupportedBy(cube_1, lid) ✓

  grasp_description=

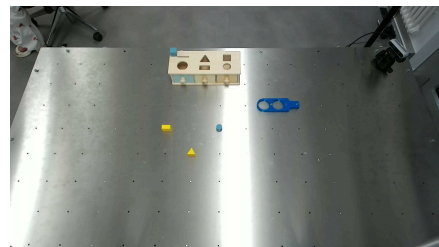
  grasp_1 = GraspDescription(
    FRONT, TOP,
    LEFT_HAND)

),
InsertionAction(
  arm=LEFT,
  object_designator=cube_1,
  grasp_description=grasp_1,
  target=

  ShapeSortingHole(
    shape_category=CUBE)

])

```



Tracy

executes the resolved plan

